MIL-C-18757/4A(OS)
15 April 1981
SUPERSEDING
MIL-C-18757/4(OS)
8 August 1974 with
Amendment 2
10 February 1976

MILITARY SPECIFICATION

CARTRIDGE CASE, STEEL, 5-INCH, MARK 9 ALL MODS (54 CALIBER)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the specific requirements for the 5"/54 Caliber Mk 9 all Mods Steel Cartridge Case, hereinafter referred to as the "cartridge case". The general requirements are covered in MIL-C-18757.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for

MILITARY

MIL-C-18757	Cartridge Cases, Steel (Other Than Small Arms) General Specification for

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

517536	5-Inch Cartridge Case Mk 9 Mod 0 (54 Caliber) Steel
2847224	Case, Cartridge Mk 9 Mod 1 5"/54 Caliber, Steel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NAVAL SEA SYSTEMS COMMAND

OD 47471 Method for Rating the Effects of Salt Spray Testing on Zinc Electroplated Cartridge Cases.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity of as directed by the contracting officer).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8-78 Tension Testing of Metallic Materials, Methods of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.) Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General requirements. The complete requirements for procuring the cartridge case described herein shall consist of MIL-C-18757 and this document.

3.2 Preproduction lot. The preproduction lot shall meet all the requirements specified herein and on Dwg 517536 or 2847224, as applicable.

3.3 Production lot. The production lot shall meet all the requirements specified herein and on Dwg 517536 or 2847224, as applicable.

3.4 Hardness. The hardness value for Region D of FIGURE 1 shall be as specified in TABLE I. Values shown for Regions A_1, A_2, B and C are advisory.

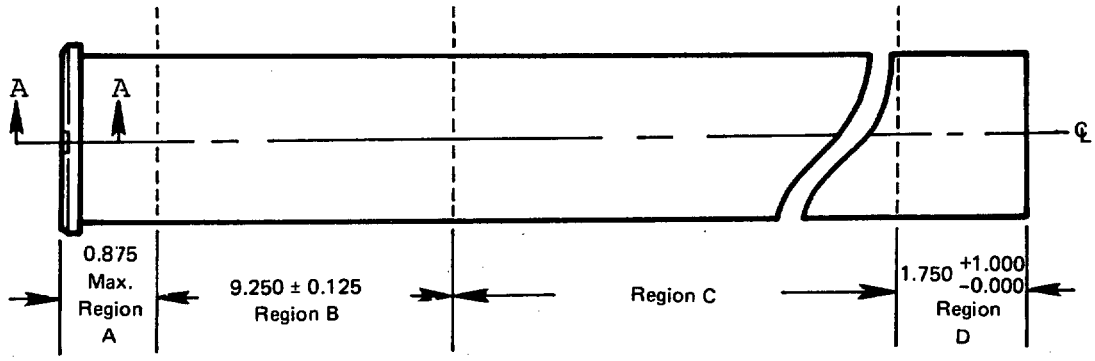
TABLE I. Hardness values. 1/

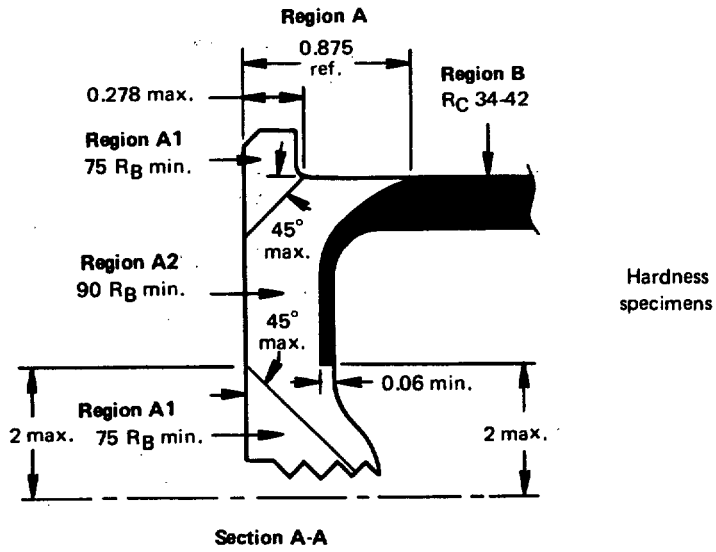
Region A_1		Region A_2		Region B		Region C		Region D
Max 2/	Min	Max 2/	Min	Max 2/	Min	Max 2/	Min	Max
42C	75B	42C	90B	42C	34C	42C	95B	85B

1/ All values given in Rockwell Scale.

2/ Paragraph 3.3.5 of QQ-Z-325B does not apply.

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Section A-A

Unless otherwise specified, all dimensions are in inches.

FIGURE 1. Hardness tensile specimen locations.

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3.5 Elongation. The elongation values shall be as specified in TABLE II.

TABLE II. Minimum elongation.

Region	Elongation in 1 inch
B	6.0%
C	5.0%

3.6 Tension. The yield strength statistical parameters, both areas, ($\bar{X}$ - mean, s - standard deviation), when polotted on FIGURES 2 and 3, must be on the k_{min} line or within the designated acceptable area. The one-sided tolerance limit factor k for the base area, when calculated, must be equal to or greater than 2.376; for the sidewall area, k_1, when calculated, must be equal to or greater than 2.823. Test samples distribution shall be as specified in TABLE III. Each test activity's (Government and contractor) test data shall be compared separately against the final acceptance requirements. (k_{min} was determined using population statistical parameters μ,σ,n,γ - see FIGURES 2 and 3).

3.7 Protective coating. The cartridge case protective coating shall meet the requirements specified on Dwg 517536 or 2847224, as applicable, and shall have a rating of the black corrosion products of zinc less than the established standard of 0.24000 when tested in accordance with 4.4.4.

4. QUALITY ASSSURACE PROVISIONS

4.1 The quality assurance provisions for the cartridge case described herein shall consist of MIL-C-18757 and this document.

4.2 Preproduction lot. The preproduction lot shall consist of 158 cartridge cases and shall be examined and tested in accordance with the Preproduction and Quality Conformance Inspection and Sampling Table of MIL-C-18757.

4.3 Production lot. Unless otherwise specified in the contract, the production lot shall consist of 5000 cartridge cases plus those required for test purposes and shall be examined and tested in accordance with the Preproduction and Quality Conformance Inspection and Sampling Table of MIL-C-18757.

4.4 Tests.

4.4.1 Hardness. Cartridge cases shall be prepared and tested in accordance with the hardness test of MIL-C-18757.

4.4.2. Elongation. Cartridge cases shall be prepared and tested in accordance with the elongation test of MIL-C-18757.

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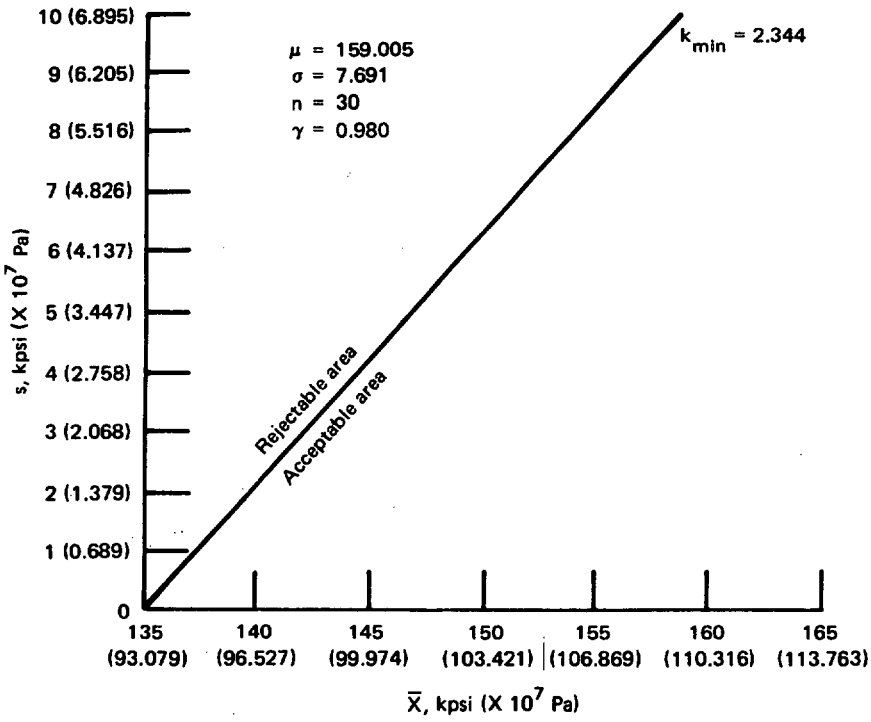


FIGURE 2. Final acceptance chart – base area.

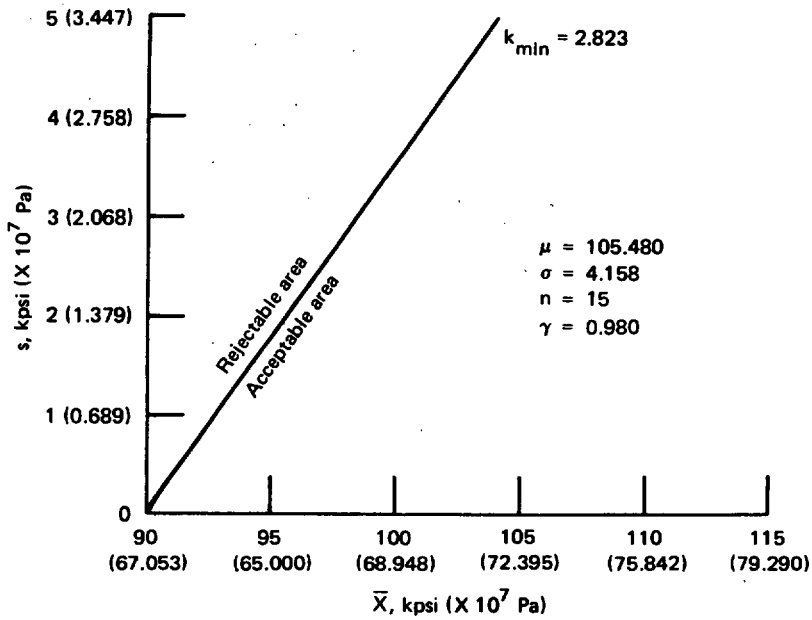
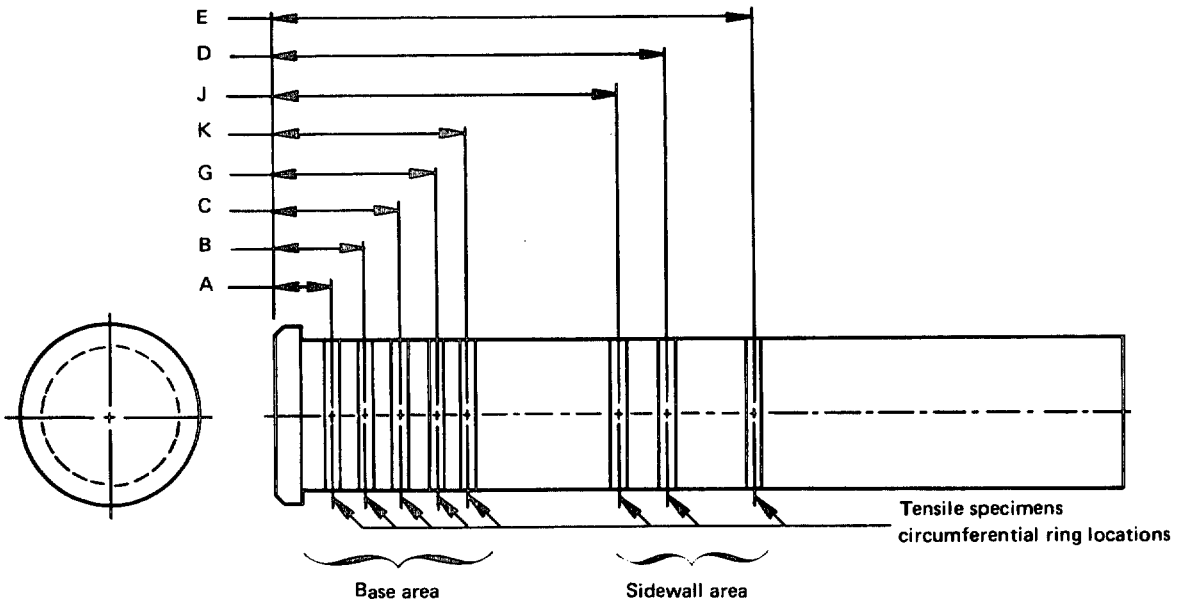


FIGURE 3. Final acceptance chart – sidewall area.

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Locations

A – 2.5 in. (6.35 cm)	K – 8.5 in. (21.59 cm)
B – 4.0 in. (10.16 cm)	J – 12.0 in. (30.48 cm)
C – 5.5 in. (13.97 cm)	D – 13.0 in. (33.02 cm)
G – 7.0 in. (17.78 cm)	E – 16.0 in. (40.64 cm)

Figure 4. Specimen location.

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TABLE III. Specimen test matrix.

Sample number	Base locations					Sidewall locations		
	A	B	C	G	K	D	E	J
1.	A	B				D		
2.	A	B					E	
3.	A	B						J
4.	A		C			D		
5.	A		C				E	
6.	A		C					J
7.	A				K	D		
8.	A			G			E	
9.	A			G				J
10.	A			G		D		
11.	A				K		E	
12.	A				K			J
13.				G	K	D		
14.		B			K		E	
15.		B	C					J
16.		B	C			D		
17.		B	C				E	
18.		B		G				J
19.		B		G		D		
20.		B		G			E	
21.		B			K			J
22.		B			K	D		
23.				G	K		E	
24.			C		K			J
25.			C	G		D		
26.			C	G			E	
27.			C	G				J
28.			C		K	D		
29.			C		K		E	
30.				G	K			J
Total	12	12	12	12	12	10	10	10

NOTES:
1. Odd-numbered samples are to be tested by contractor and the even-numbered samples are to be tested by Government test activity, WPNSTA Seal Beach.

2. Samples must be randomly selected from submitted lot.

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4.4.3 Tension.

4.4.3.1 Specimen preparation. Cut and identify circumferential rings from 30 cases (per lot) in accordance with specimen location sketch (FIGURE 4) and matrix (TABLE III). Maintain ring widths slightly greater than the final specimen grip width to allow for subsequent shaping operations. Prior to cutting the rings, specimen segments A, B, and C will require surface machining (both inside and outside surfaces) to produce parallel specimen face surfaces (nominal thickness to be between 0.060 and 0.080 inches). Specimen ring segments G, K, J, D, and E will not require surface machining prior to ring cutting. Cut the rings in half, each half to have identical identification. Retain one set of ring segments for possible retest. Remove edge burrs and sand edges slightly to remove any possible stress raisers. Straighten specimens by passing them through a sheet metal roller, using a minimum of passes, until bowing cannot be detected with a straight edge. Machine all specimen segments to dimensions specified in ASTM E8, FIGURE 6, subsize specimens. Several specimens may be machined at one time. Individually, sand the edges of the gauge section to obtain a 0.003 - 0.005 inch taper from the end to the middle of the gauge section. Smooth all edges of the necked down area. Stress relieve all segments at 610 ± 10°F for 30 ± 5 minutes and then cool to room temperature for a minimum of one hour.

4.4.3.2 Specimen testing. Perform hardness tests on all specimens by taking measurements in the specimen grip area or adjacent to the grip area away from the necked down area. Record all measurements. Perform hardness tests in accordance with the hardness test of MIL-C-18757. Perform tensile and elongation tests on all specimens in accordance with the tensile and elongation tests of MIL-C-18757. Test all tensile specimens using a head separation rate of 0.05 inches per minute. Retain original stress-strain curves. Note magnification and scale on all stress-strain curves. Measure gauge dimensions as accurately as possible. Record gauge cross sectional area to four decimal places.

4.4.3.3 Acceptance procedure.

4.4.3.3.1 Determine lot mean (both areas) of yield strength test values using mean equation, $\bar{X} = \frac{\Sigma X}{n}$, where X = test value and n = sample size.

4.4.3.3.2 Determine lot standard deviations (both areas) of yield strength test values using standard deviation equation, $s = \left[\frac{n\Sigma X^2 - (\Sigma X)^2}{n(n-1)}\right]^{1/2}$

4.4.3.3.3 Plot sample $\bar{X}$ and s on yield strength final acceptance charts (FIGURES 2 and 3) or calculate lot k value from equation, $k = \frac{\bar{X} - X_L}{s}$, where $\bar{X}$ = lot mean, X_L = lower limit yield value (base = 135,000 psi, sidewall = 90,000 psi), s = lot standard deviation and k = one-sided tolerance limit factor.

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4.4.4 Protective coating. Cartridge cases shall be prepared and tested in accordance with the Protective Coating Test of MIL-C-18757. The assigned plating thickness value of the interior recessed area of the cartridge case head shall be determined as follows:

4.4.4.1 Three specimen sections shall be taken from each case, at approximately 120° apart, and six measurements, approximately 0.1 inch apart, shall be taken on each segment (see FIGURE 5).

4.4.4.2 The assigned plating thickness value (T) shall be calculated as follows:

T = plating thickness, any case
t_a = plating thickness, segment a
t_b = plating thickness, segment b
t_c = plating thickness, segment c
t_{a1} = plating thickness reading preceding t_{a2}
t_{a2} = minimum plating thickness reading of segment a
t_{a3} = plating thickness reading following t_{a2}

$$t_a = \frac{t_{a1} + 6\,t_{a2} + t_{a3}}{8}$$

$$T = \frac{t_a + t_b + t_c}{3}$$

t_b and t_c are computed in a similar manner

NOTE: (1) If t_{a2} is a measurement at location 1 or 6, use $t_{a1} = t_{a3}$ in above formula (use on any segment). See FIGURE 5.

(2) Record all thickness readings.

The black speck corrosion rating prodecure shall be as specified in OD 47471.

5. PACKAGING

5.1 Preparation for delivery shall be in accordance with MIL-C-18757 and as specified herein.

5.2 Packing. Packing shall be in accordance with MIL-C-18757 and the following:

a. Four (4) cartridge cases shall be in each container.

b. Box dimensions shall be 12 11/16 x 12 11/16 x 33 1/8 inches.

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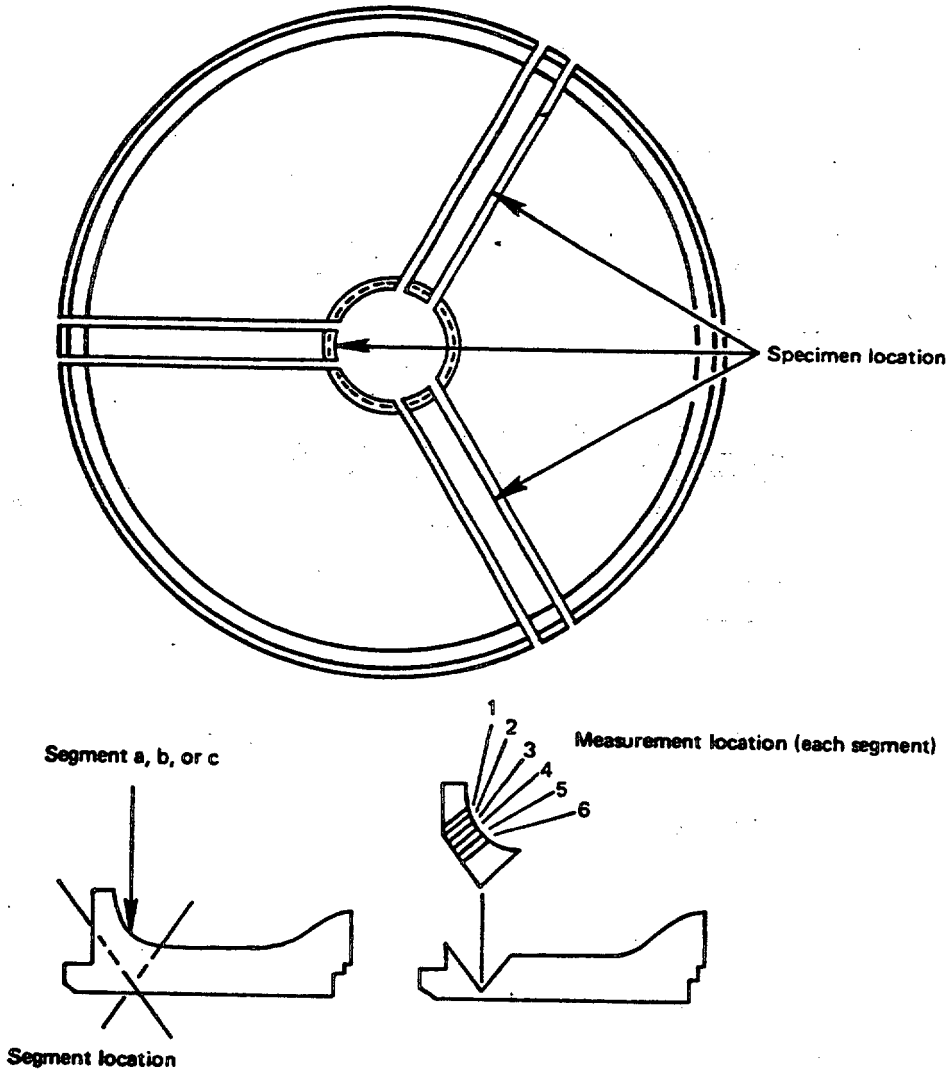


FIGURE 5. Metallographic sectioning.

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c. Partition dimensions shall be 37 7/8 x 12 5/8 inches, two required per container, one center slotted 22 1/8 inches and one center slotted 15 13/16 inches for assembly, two fold-out flaps 6 5/16 inches per partition for bottom pad.

d. Top pad dimensions shall be 15 9/16 x 12 9/16 inches with 1 1/2 inch flange with four holes 5 3/8 inches in diameter.

Note: Tolerances for all packing dimensions shall be ±1/32 inch.

6. NOTES

6.1 Intended use. The 5"/54 caliber Mk 9, all mods, cartridge cases, when assembled with their appropriate primers, propellant charges, and plugs, and when used with their appropriate loaded projectiles, form complete rounds which are used as service ammunition in the 5"/54 caliber gun.

6.2 Ordering data. Ordering data shall be in accordance with MIL-C-18757.

6.2.1 Contractor's tests. Tests on odd numbered specimens from TABLE III for hardness, elongation, and tension shall be conducted by the contractor at the contractor's site.

6.2.2 Government tests. Tests on even numbered specimens from TABLE III for hardness, elongation, and tension shall be conducted by the Government at a facility designated by the procuring activity.

6.3 Statistical glossary.

k	One-sided tolerance limit factor
n	Sample size
s	Sample standard deviation
X	Test value
$\overline{X}$	Sample mean (average)
X_L	Minimum yield strength
γ	Confidence level
μ	Population mean
σ	Population standard deviation
Σ	Summation

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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